Chapter -II- Connected Graphs

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Pristini School of AI
Academic year 2023 - 2024

10 mars 2024

1 Path and circuit (case of a directed graph)
- Eulerian path

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2. Chain and cycle (Undirected graph)

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2. Chain and cycle (Undirected graph)
3. Connectedness (Undirected graph)
4. Strong connectedness (directed graph)

Path (directed graph)

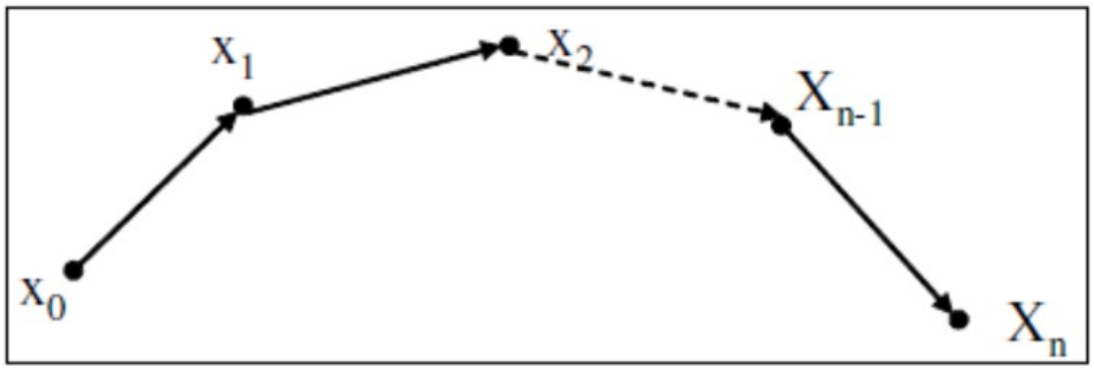
- A **path** is a sequence of arcs $(u_1, u_2, u_3, \ldots, u_n)$ such that $I(u_{i+1}) = T(u_i) \quad \forall \quad 1 \leq i \leq n-1$.

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- In other words : a path is a succession of arcs walked in the same direction.

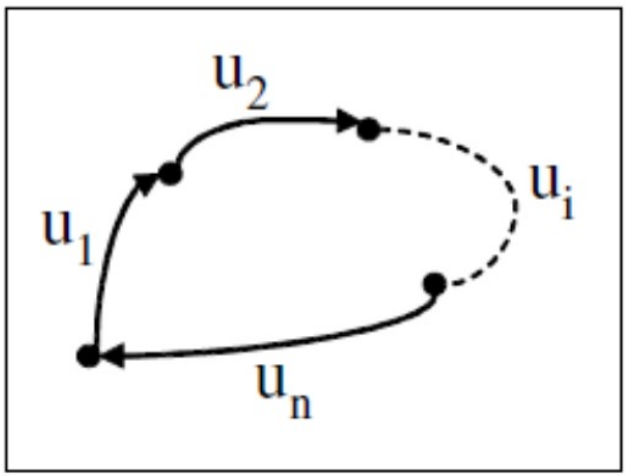
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- Example :



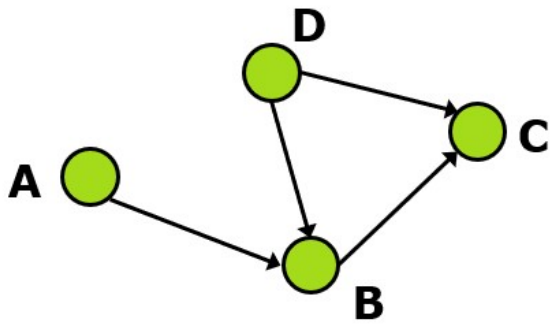
Circuit (directed graph)

1. A **circuit** is a path that begins and ends at the same vertex.
2. A **circuit** is a path such that : $T(u_n) = I(u_1)$.



Path & Circuit (directed graph)

Example

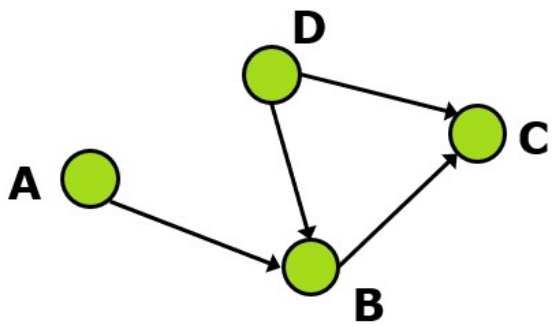


1. Is there a path from A to D ?

Solution

Path & Circuit (directed graph)

Example

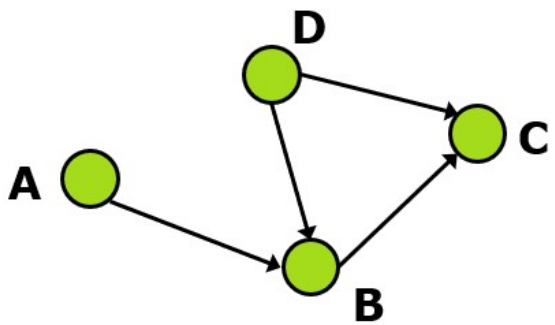


1. Is there a path from A to D ?
2. Does the graph contain any circuit ?

Solution

Path & Circuit (directed graph)

Example



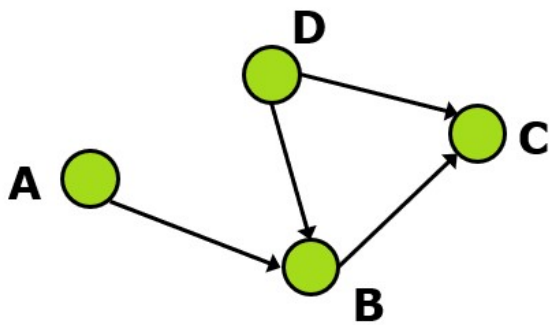
1. Is there a path from A to D ?
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Solution

1. No

Path & Circuit (directed graph)

Example



1. Is there a path from A to D ?
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Solution

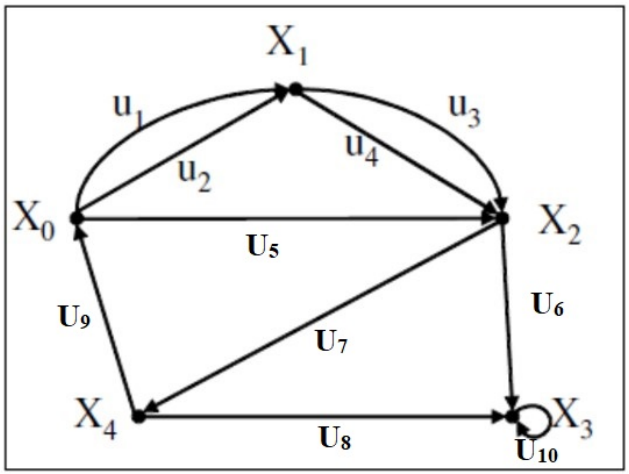
1. No
2. No

- **Length :** The **path** length corresponds to the number of arcs in the path.

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- **Distance :** The distance between two **vertices** of a graph is the length of the shortest path between these two vertices.

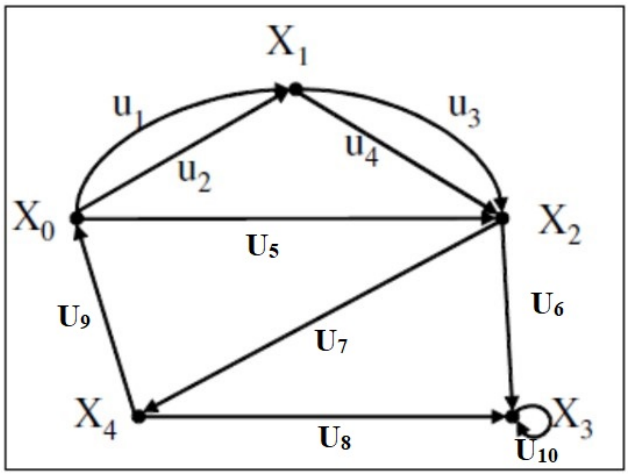
- **Length :** The **path** length corresponds to the number of arcs in the path.
- **Distance :** The distance between two **vertices** of a graph is the length of the shortest path between these two vertices.
- **Diameter :** The diameter of a graph is the **maximum distance** which separates any two vertices of this graph.

Exercise : Consider the following graph G



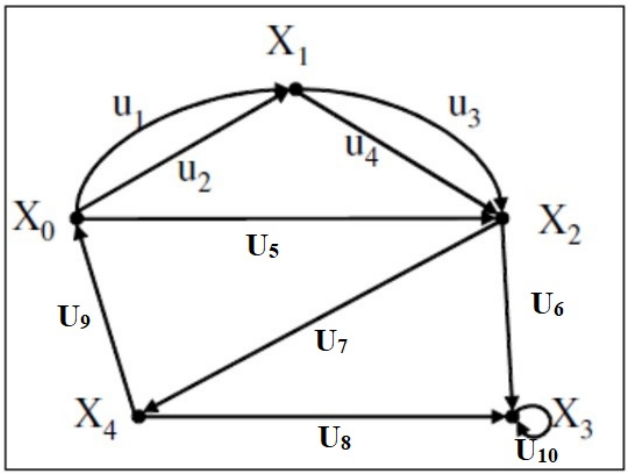
1. Determine the possible paths between X_0 and X_3 as well as their lengths ?

Exercise : Consider the following graph G



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2. Does graph G contain a circuit ?

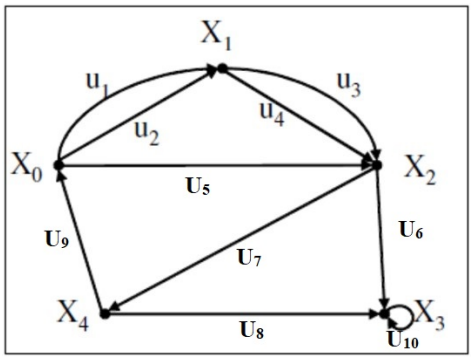
Exercise : Consider the following graph G



1. Determine the possible paths between X_0 and X_3 as well as their lengths ?
2. Does graph G contain a circuit ?
3. What is the diameter of graph G ?

Different types of paths (directed graph)

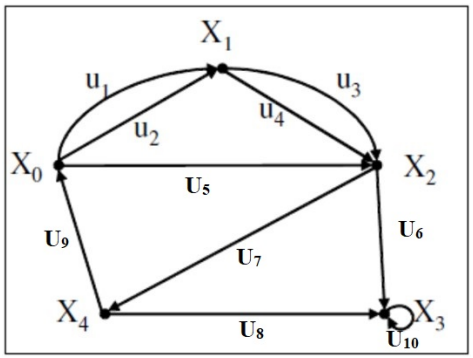
Solution :



1.

Different types of paths (directed graph)

Solution :

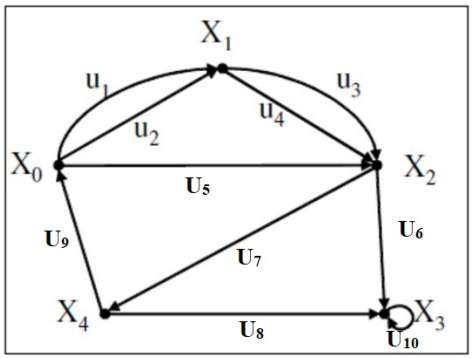


1.

2.

Different types of paths (directed graph)

Solution :



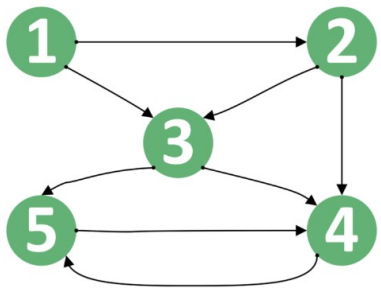
1.
2.
3.

Different types of paths (directed graph)

A) Simple path

A simple path is a path in a graph which does not have repeating arcs.

Example



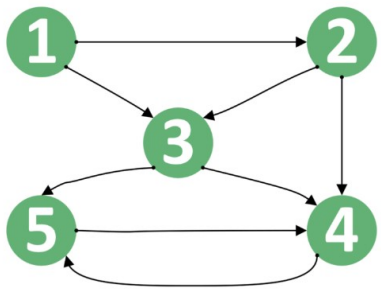
1. How many simple paths between vertices 1 and 4 are in graph G ?

Different types of paths (directed graph)

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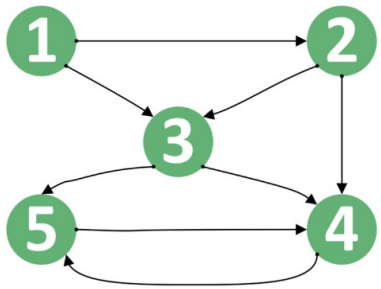
Example



1. How many simple paths between vertices 1 and 4 are in graph G ?
2. Is $(1, 3, 4, 5, 4)$ a simple path ? Justify your answer.

Different types of paths (directed graph)

A) Simple path

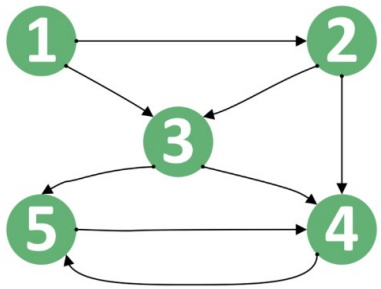


Answer

1. According to graph G, there are 10 simple paths between vertices 1 and 4 : $P_1 = (1, 2, 3, 4)$, $P_2 = (1, 2, 3, 4, 5)$; $P_3 = (1, 2, 3, 5)$; $P_4 = (1, 2, 3, 5, 4)$, $P_5 = (1, 2, 4)$; $P_6 = (1, 3, 4)$; $P_7 = (1, 3, 5)$; $P_8 = (1, 3, 5, 4)$; $P_9 = (1, 2, 4, 5)$; $P_{10} = (1, 3, 4, 5)$

Different types of paths (directed graph)

A) Simple path



Answer

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2. The path $P = (1, 3, 4, 5, 4)$ is not simple because it contains a cycle - vertex 4 appears two times in the sequence.

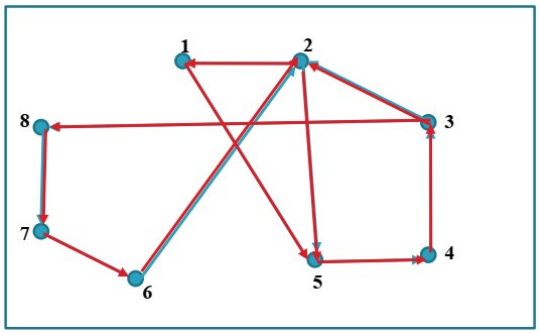
Different types of paths (directed graph)

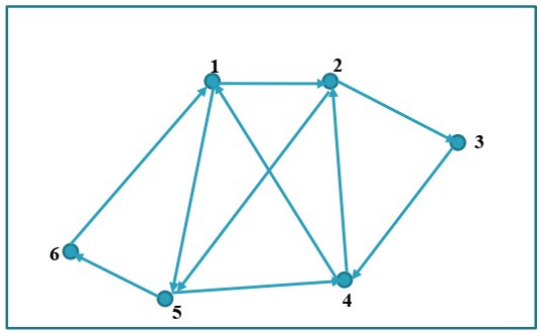
B) **Elementary path**
A path is said to be elementary if it does not visit the same vertex several times.

Different types of paths (directed graph)

c) **Eulerian path**

A path is called Eulerian path if it visits every edge exactly once. This definition works for both **directed and undirected graphs**.





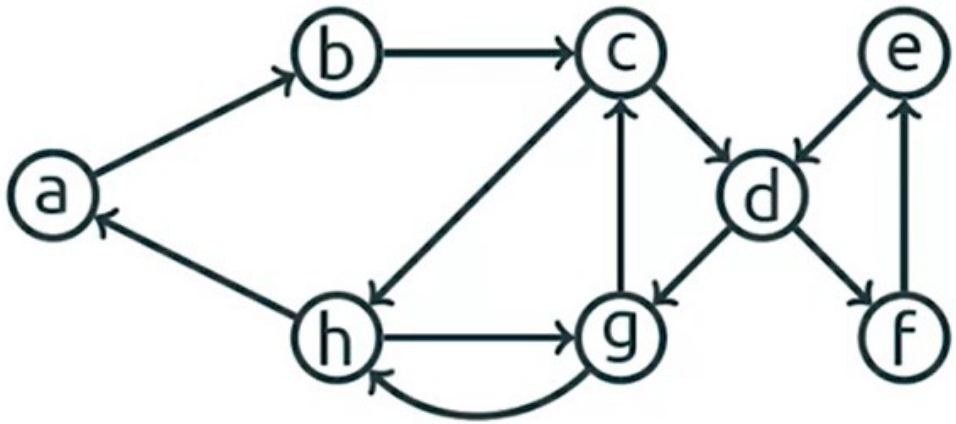
Different types of paths (directed graph)

Theorem

A connected *undirected* graph contains an Eulerian cycle, if and only if the degree of every vertex is even.

Different types of paths (directed graph)

Proof (Directed Case)

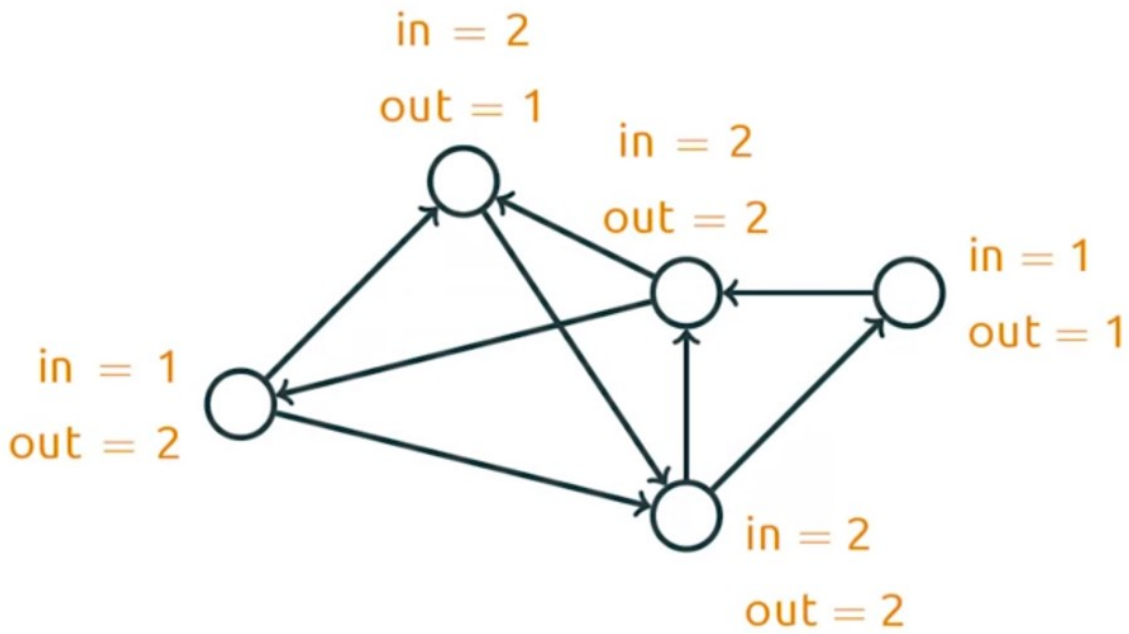


if some node is imbalanced,
there is clearly no Eulerian cycle

Different types of paths (directed graph)

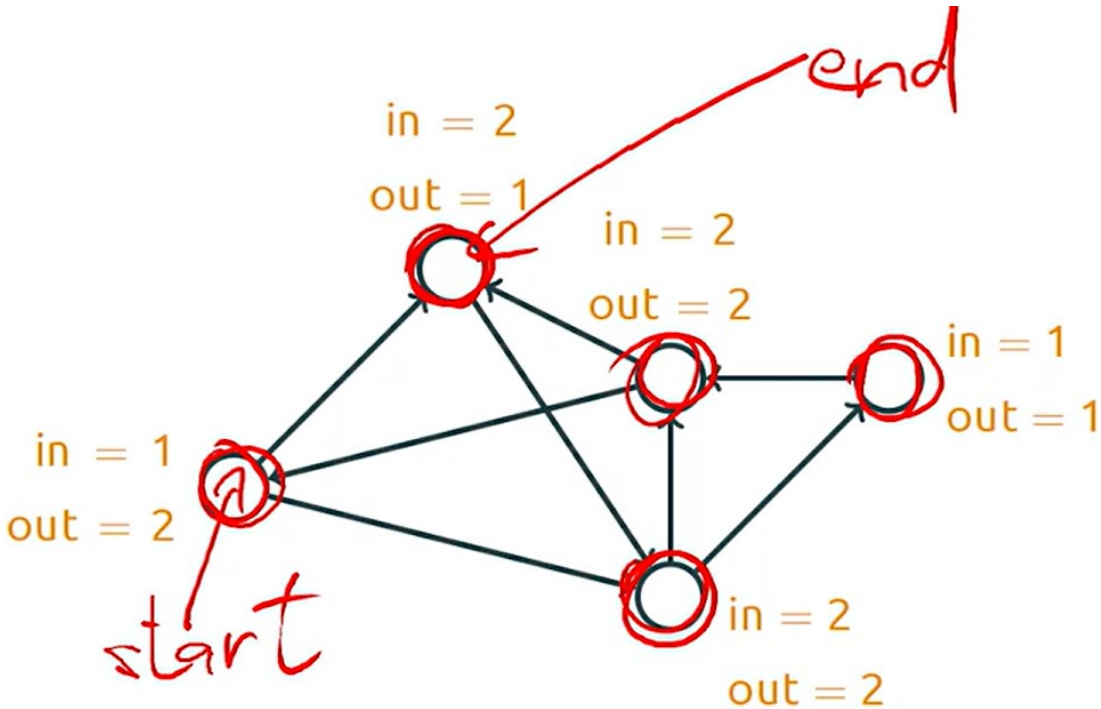
From path to cycle

- Is there an Eulerian path in this graph ?



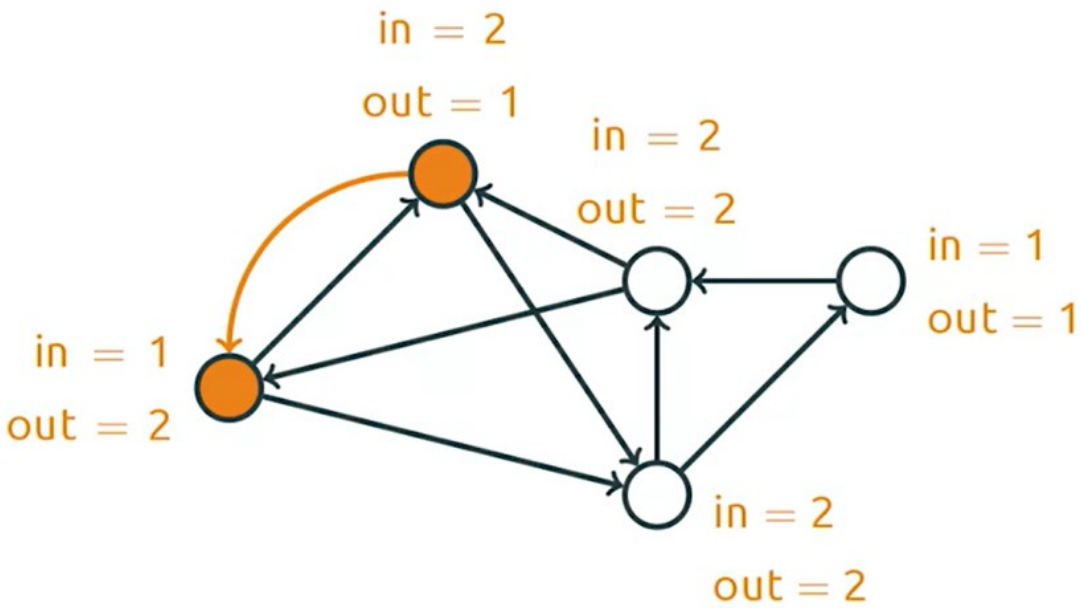
Different types of paths (directed graph)

From path to cycle



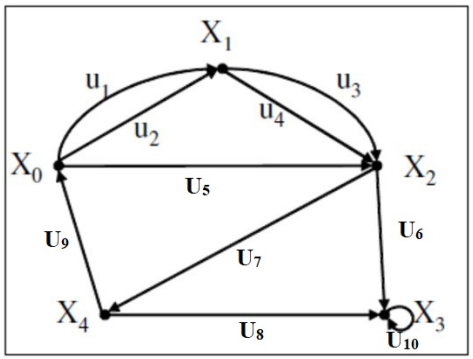
Different types of paths (directed graph)

From path to cycle



Different types of paths (directed graph)

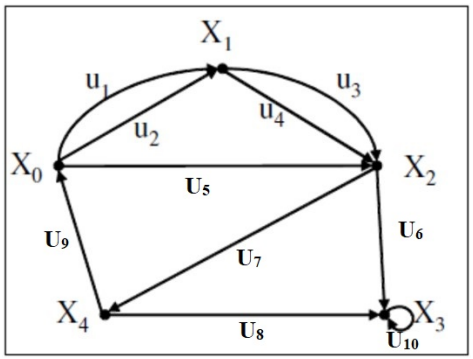
Example



1. The path : (U_1, U_3, U_6) is a simple path.

Different types of paths (directed graph)

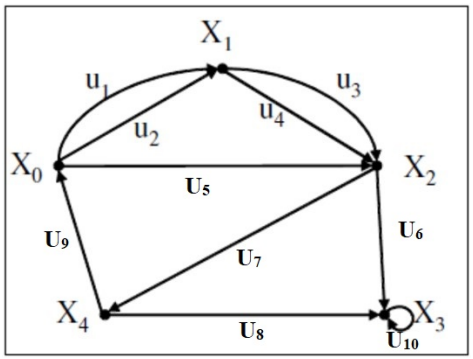
Example



1. The path : (U_1, U_3, U_6) is a simple path.
2. The path : $(U_5, U_7, U_9, U_2, U_4, U_6)$ is a simple path.

Different types of paths (directed graph)

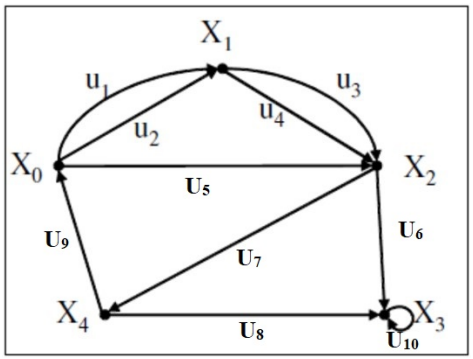
Example



1. The path : (U_1, U_3, U_6) is a simple path.
2. The path : $(U_5, U_7, U_9, U_2, U_4, U_6)$ is a simple path.
3. The path : (U_1, U_3, U_6) is an elementary path.

Different types of paths (directed graph)

Example



1. The path : (U_1, U_3, U_6) is a simple path.
2. The path : $(U_5, U_7, U_9, U_2, U_4, U_6)$ is a simple path.
3. The path : (U_1, U_3, U_6) is an elementary path.
4. The path : $(U_5, U_7, U_9, U_2, U_4, U_6)$ is a non-elementary path

Different types of paths (directed graph)

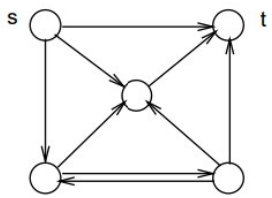
D) **Hamiltonian path**

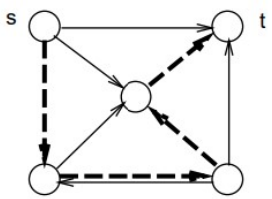
A Hamiltonian path is a directed path that goes through each node exactly once. In other words, a path is Hamiltonian if it is elementary and visits all vertices.

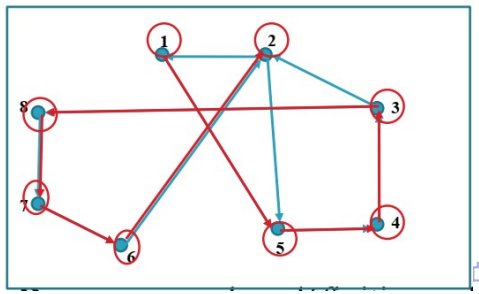
Example Given a directed graph G and vertices s and t in this graph, is there a Hamiltonian path from s to t in G ?

HAMPATH=

$\{\langle G, s, t\rangle / G$is a directed graph having a Hamiltonian path from s to $t\}$

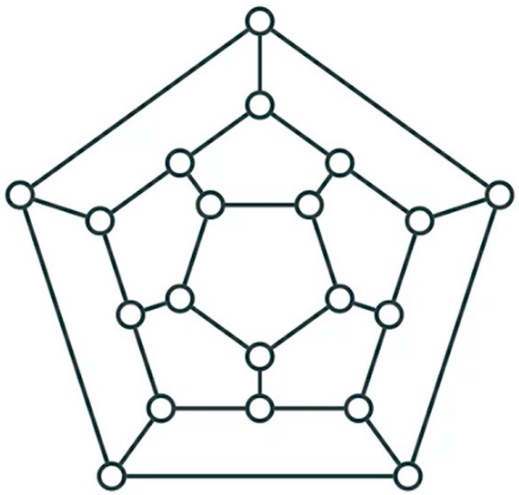






A **Hamiltonian cycle** visits every node of a graph exactly once.

Example



Summary

- Eulerian cycle visits every **edge** exactly once.
- Hamiltonian cycle visits every **node** exactly once.
- Look similar to each other, but differ drastically from the computational point of view.

Important Note : the definitions of simple, elementary, Eulerian and Hamiltonian **circuits** is analogous to those of **paths**.

Chain and cycle (Undirected graph)

By analogy with paths and circuits in the case of a **directed** graph, we define **chains** and **cycles** using the following vocabulary :

Directed graph	Undirected graph
Arc	Edge
Path	Chain
Circuit	Cycle

Connectedness (Undirected graph)

- We define the **binary connection relation** C on the set of **vertices** V of a graph G by :

$$xCy \Leftrightarrow \exists \quad \text{a chain between} \quad x \quad \text{and} \quad y.$$

C is an **equivalence relation**. In fact, C is :

- ◇ Reflexive : xCx
- ◇ Symmetric : $xCy \Leftrightarrow yCx$
- ◇ Transitive :

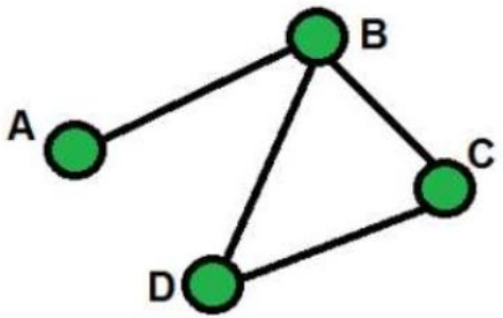
$$\begin{cases} xCy \\ yCz \end{cases} \Leftrightarrow xCz$$

- The equivalence classes induced on X by this relation form a partition of X into x_1, x_2,...,x_p.
- The number p of distinct equivalence classes is called the connectivity number of the graph.

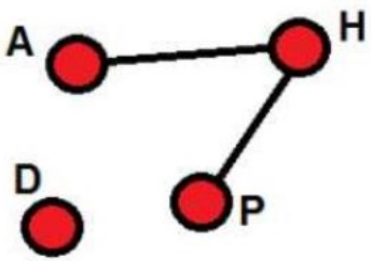
Connectedness (Undirected graph)

- Definition

A graph is said to be **connected**, if for every pair of vertices i and j, there exists a **chain** joining i and j.



Connected graph

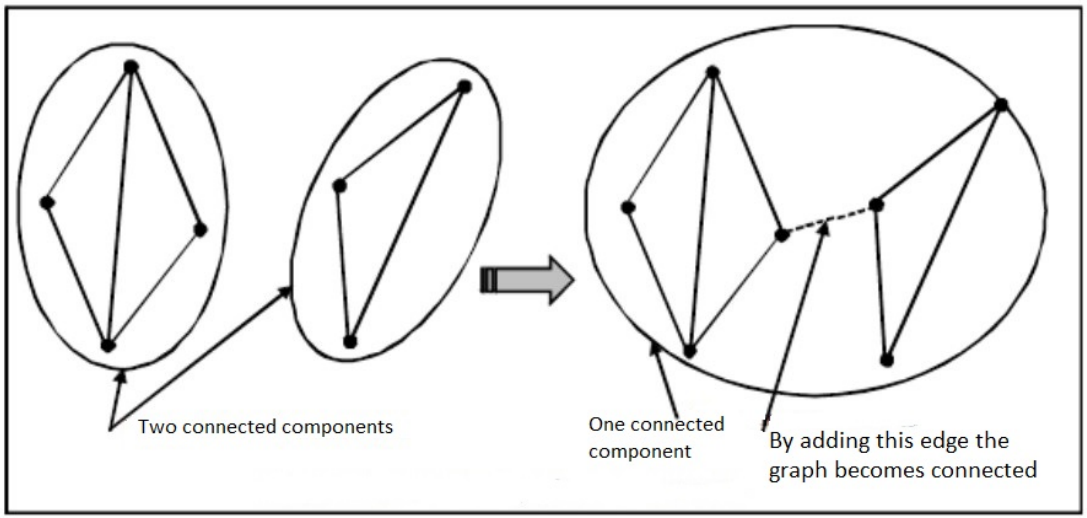


Non-connected graph

Connectedness (Undirected graph)

Connected components of a graph

- A graph G is said to be connected if the number of its connected components is equal to 1.
- The subgraphs G_i generated by the subsets x_i of vertices, are called the connected components of the graph. Each connected component is a connected graph.
- **Example** :



Strong connectedness (directed graph)

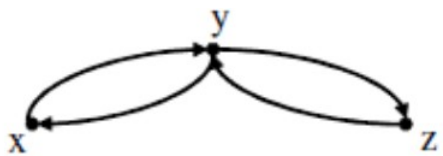
Connected components of a graph

- We define the **binary connection relation** $\widehat{C}$ on the set of **vertices** V of a graph G by :

$$x\widehat{C}y \Leftrightarrow \begin{cases} \text{it exists a path from} x \text{ to } y \\ \text{it exists a path from} y \text{ to } x \end{cases}$$

- This relationship of strong connectivity is an **equivalence relation**. In fact, $\widehat{C}$ is :
  - ◇ Reflexive : $x\widehat{C}x$
  - ◇ Symmetric : $x\widehat{C}y \Leftrightarrow y\widehat{C}x$
  - ◇ Transitive :

$$\begin{cases} x\widehat{C}y \\ y\widehat{C}z \end{cases} \Leftrightarrow x\widehat{C}z$$



Strong connectedness (directed graph)

Theorem

A strongly connected *directed* graph contains an Eulerian cycle, if and only if, for every vertex, its in-degree is equal to its out-degree.

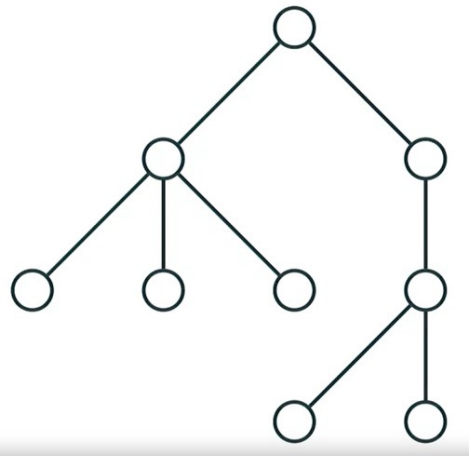
Trees

Definitions

1. A **tree** is a connected graph without cycles
2. A **tree** is a connected graph on n vertices and $n-1$ edges
3. A graph is a tree if and only if there is a unique simple path between every pair of its vertices

Trees

Example



- It's quite easy to see that this graph is connected and doesn't have any cycles. So it is a tree according to the first definition.
- This is a connected graph on nine vertices and eight edges, which means it is a tree according to the second definition as well.
- For the third definition, we want to show that there exists a unique simple path between every pair of vertices. I want to show you, for every pair of vertices, I'll show you for one pair.

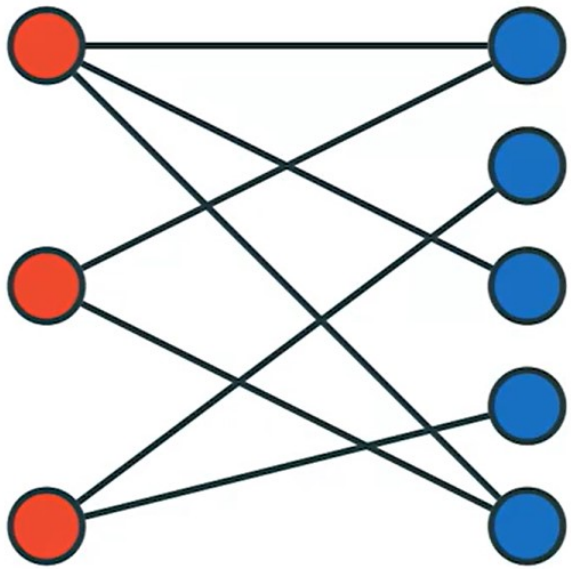
Bipartite graphs

Definitions

1. A graph G is **bipartite** is its vertices can be partitioned into **two disjoint sets** L and R such that
   - Every edge of G connects a vertex in R to a vertex in L.
   - i.e no edge connects tow vertices from the same part
2. L and R are called the **parts** of G

Bipartite graphs

Example



Bipartite graphs : Characterization

Theorem

A graph is **bipartite**, if and only if, it has no cycles of odd length.

Depth First Search (DFS)

Depth first Search or Depth first traversal is a recursive algorithm for searching all the vertices of a graph or tree data structure. Traversal means visiting all the nodes of a graph.

Depth First Search Algorithm

A standard DFS implementation puts each vertex of the graph into one of two categories :

1. Visited
2. Not Visited

The purpose of the algorithm is to mark each vertex as visited while avoiding cycles.

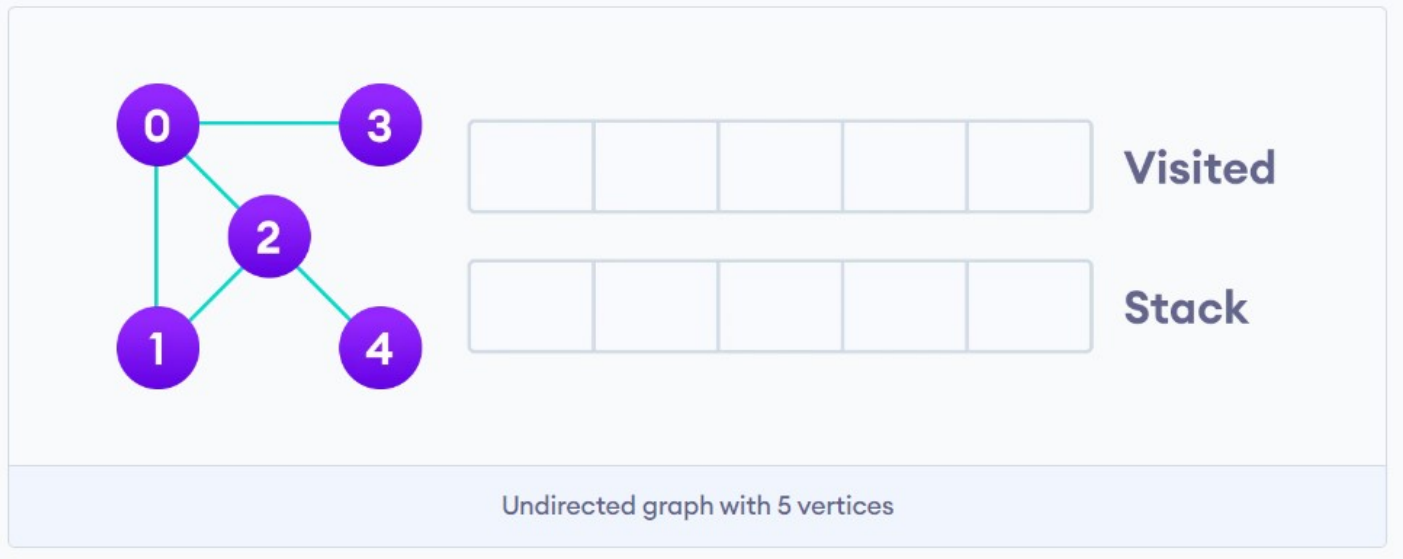
Depth First Search Example

The DFS algorithm works as follows :

1. Start by putting any one of the graph's vertices on top of a stack.
2. Take the top item of the stack and add it to the visited list.
3. Create a list of that vertex's adjacent nodes. Add the ones which aren't in the visited list to the top of the stack.
4. Keep repeating steps 2 and 3 until the stack is empty.

Depth First Search Example

The DFS algorithm works as follows :



Depth First Search Example

The DFS algorithm works as follows :We start from vertex 0, the DFS algorithm starts by putting it in the Visited list and putting all its adjacent vertices in the stack.



Depth First Search Example

Next, we visit the element at the top of stack i.e. 1 and go to its adjacent nodes. Since 0 has already been visited, we visit 2 instead.



Depth First Search Example

Vertex 2 has an unvisited adjacent vertex in 4, so we add that to the top of the stack and visit it.

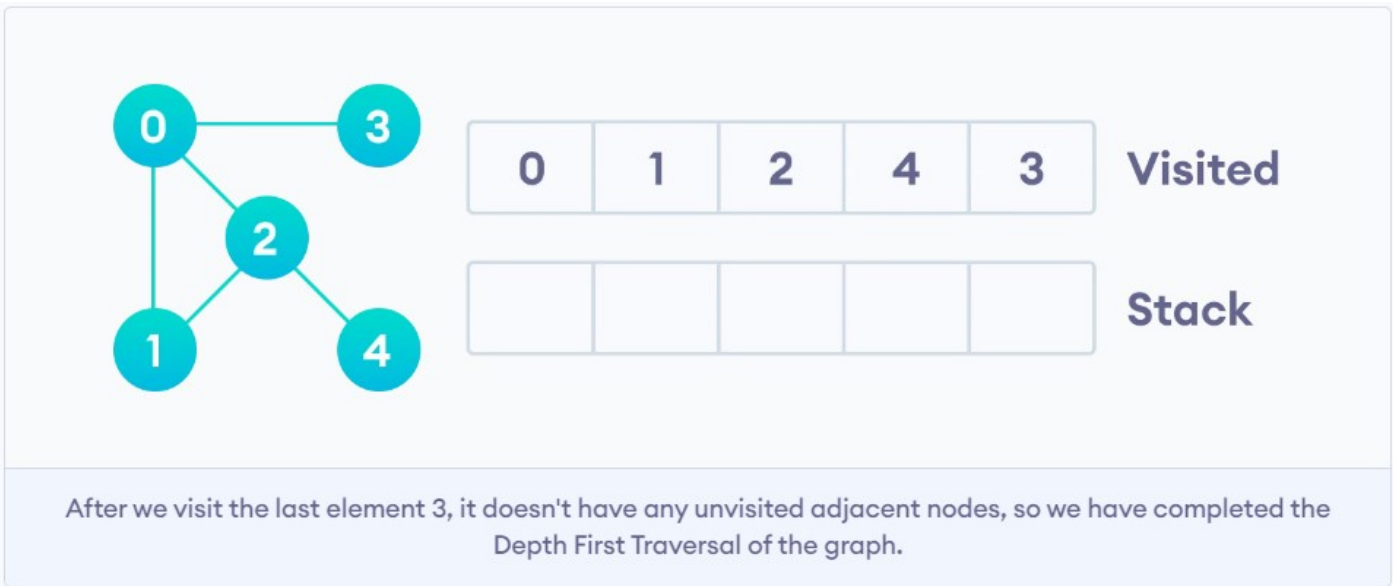


Depth First Search Example



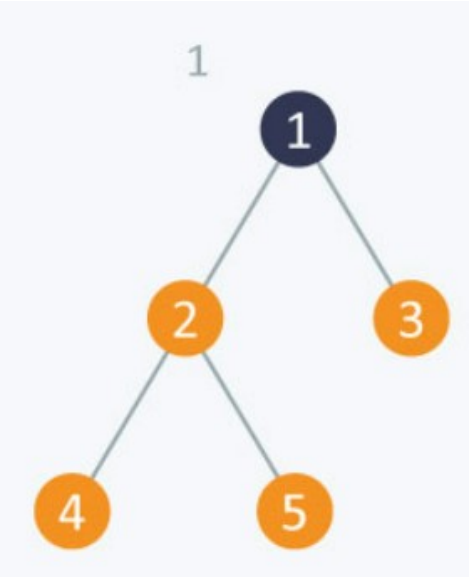
Depth First Search Example

After we visit the last element 3, it doesn't have any unvisited adjacent nodes, so we have completed the Depth First Traversal of the graph.

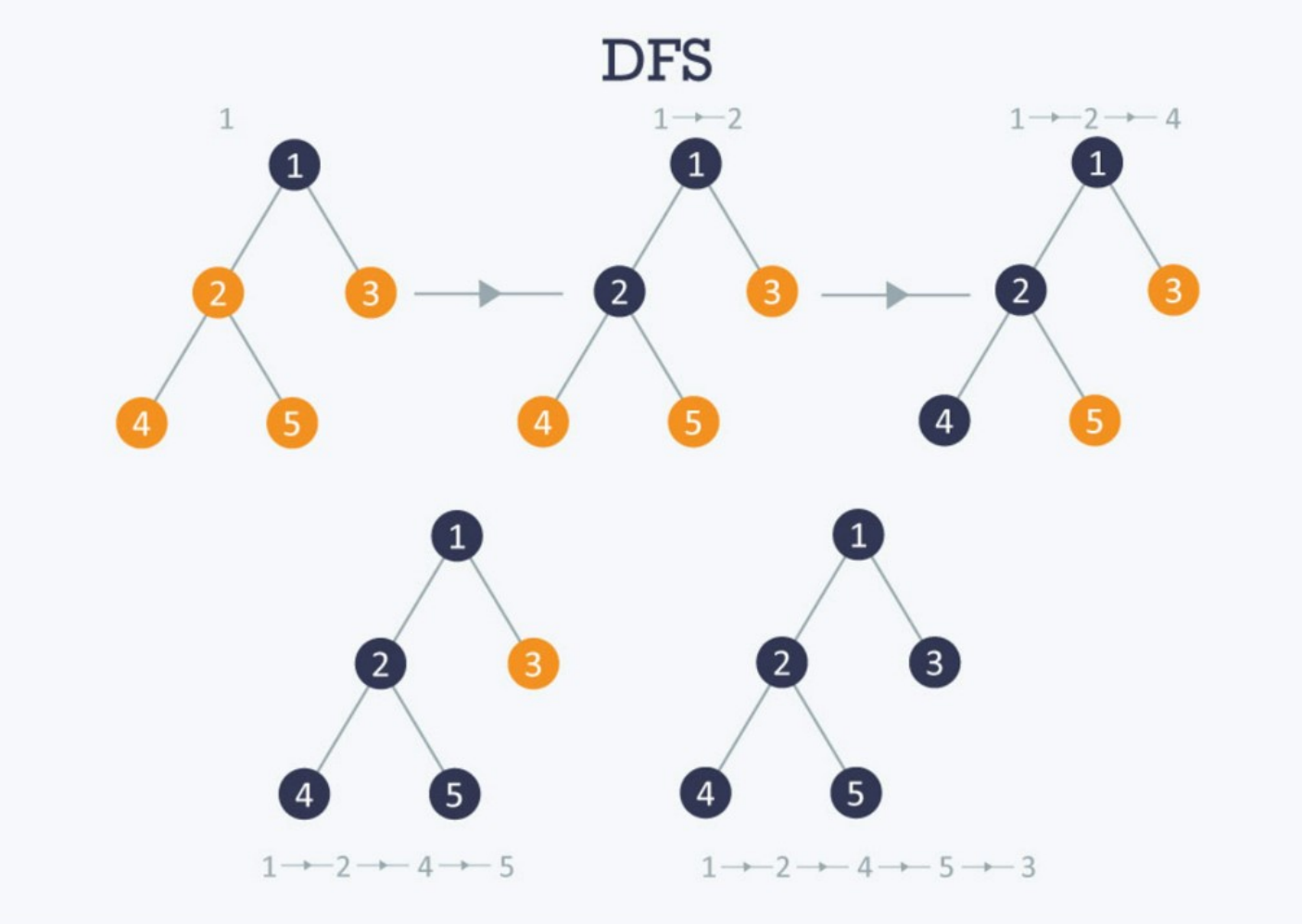


Conclusion : $DFS = \{0, 1, 2, 4, 3\}$

Depth First Search - Exercise



Depth First Search - Exercise



References

- *Wilson R.J., (2010), Introduction to Graph Theory, Fifth edition, Pearson*
- *Diestel R., (2017), Graph Theory, Fifth edition, Springer Nature*
- *Balakrishnan V.K., (1997), Theory and problems of Graph Theory, Shhaum's outline series, McGraw-HILL.*
- *Introduction to Graph Theory, University of California San Diego, (20 hours), COURSERA.*